WORKED SOLUTION

STEP II 1996, Question 1

Pure • Binomial Series Coefficients

STEP II 1996, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/96-s2-q1

The key idea

A coefficient in a product of three brackets is a sum over all ways of choosing one term from each bracket so that the powers add up to the target. In (i) that means organising a finite count carefully. In (ii) the coefficients $1, -2, 3, -4, \ldots$ are recognisable as those of $(1+x)^{-2}$, and the extra terms in the bracket are exactly the ones needed to make the truncation invisible at x^6.

Solution

Part (i)

Write the bracket as $P(x) = \sum_{k=0}^{4} a_k x^k$ with

$$a_k = (-1)^k(k+1), \qquad k = 0, 1, 2, 3, 4.$$

Choosing $a_p x^p$, $a_q x^q$, $a_r x^r$ from the three factors contributes to x^6 exactly when $p+q+r=6$, so the coefficient we want is

$$\sum_{\substack{p+q+r=6 \\ 0\le p,q,r\le 4}} a_p a_q a_r.$$

Every term has sign $(-1)^{p+q+r} = (-1)^6 = +1$, so we may ignore signs and add up the products $(p+1)(q+1)(r+1)$.

Ask yourself How do we make sure no choice of exponents is missed or counted twice?

List the unordered triples $\{p, q, r\}$ with sum 6 and every entry at most 4, then multiply by the number of orders in which each can be taken from the three (distinguishable) brackets: 6 orders if the entries are all different, 3 if exactly two are equal, 1 if all three are equal.

exponents	product $(p+1)(q+1)(r+1)$	orders	contribution
$4,2,0$	$5\cdot 3\cdot 1 = 15$	6	90
$4,1,1$	$5\cdot 2\cdot 2 = 20$	3	60
$3,3,0$	$4\cdot 4\cdot 1 = 16$	3	48
$3,2,1$	$4\cdot 3\cdot 2 = 24$	6	144
$2,2,2$	$3\cdot 3\cdot 3 = 27$	1	27

These are all the possibilities: the largest entry is 4, 3 or 2 (it can't be less, since $3\times 1 < 6$, and it can't exceed 4), and for each largest entry we have listed every way of splitting the remainder into two smaller parts. Adding,

$$90 + 60 + 48 + 144 + 27 = 369,$$

so the coefficient of x^6 is $\boxed{369}$.

Part (ii)

Now the bracket is $Q(x) = \sum_{k=0}^{6}(-1)^k(k+1)x^k$. The binomial expansion, valid for $|x|<1$, gives

$$(1+x)^{-2} = \sum_{k=0}^{\infty}\binom{-2}{k}x^k = \sum_{k=0}^{\infty}(-1)^k(k+1)x^k,$$

since $\binom{-2}{k} = \frac{(-2)(-3)\cdots(-k-1)}{k!} = (-1)^k(k+1)$. So $Q(x)$ is the expansion of $(1+x)^{-2}$ cut off after the x^6 term:

$$(1+x)^{-2} = Q(x) + x^7R(x)$$

for some power series $R(x)$.

Ask yourself Why doesn't the missing tail $x^7R(x)$ affect the coefficient of x^6 in the cube?

Cubing,

$$(1+x)^{-6} = Q(x)^3 + 3Q(x)^2\,x^7R(x) + 3Q(x)\,x^{14}R(x)^2 + x^{21}R(x)^3.$$

Every term after the first is x^7 times a power series, so contains no power of x below x^7. Hence the coefficient of x^6 in $Q(x)^3$ equals the coefficient of x^6 in $(1+x)^{-6}$. (Put another way: a product of exponents adding to 6 can never use a term of degree 7 or more from any factor.)

By the binomial expansion again,

$$\binom{-6}{6} = \frac{(-6)(-7)(-8)(-9)(-10)(-11)}{6!} = \frac{6\cdot 7\cdot 8\cdot 9\cdot 10\cdot 11}{720} = \binom{11}{5} = 462.$$

So the coefficient of x^6 in $Q(x)^3$ is $\boxed{462}$.

Checks

Part (ii) gives an independent check on part (i). The two brackets differ only in the terms $-6x^5$ and $7x^6$, so $462 - 369$ should be the total of all contributions that use one of them. A term using $7x^6$ must take 1 from the other two brackets: 3 positions, each giving 7, total 21. A term using $-6x^5$ must take $-2x$ and 1 from the other two: $3 \times 2 = 6$ arrangements, each giving $(-6)(-2)(1) = 12$, total 72. Indeed $21 + 72 = 93 = 462 - 369$. Multiplying out the cubes by computer gives 369 and 462 directly.

Common slips

- Forgetting to count orders: the exponents $4, 2, 0$ can come from the three brackets in 6 ways, not 1.
- Including triples with an entry of 5 or 6 in part (i), where the bracket stops at x^4.
- Worrying about signs term by term: since the exponents always add to 6, every contribution is positive.

Beyond STEP

The argument in (ii) is the start of the theory of *formal power series*: series can be multiplied coefficient by coefficient without any thought of convergence, and the coefficient of x^n in a product depends only on the terms up to x^n in each factor. Treating a sequence through its generating function in this way is a powerful counting tool; for example, the coefficient of x^n in $(1 - x)^{-m}$ counts the ways of writing n as an ordered sum of m non-negative integers. Siklos's *Advanced Problems in Mathematics* has more STEP-style practice with expansions of this kind.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.